MIL-I-13548A
16 March 1966
SUPERSEDING
MIL-I-13548 (SigC)
20 July 1954

MILITARY SPECIFICATION

INSULATION SLEEVING, ELECTRICAL

(CABLE SPLICING)

FT-96, FT-97, FT-98, AND FT-472

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers four types of impregnated, punched, cotton sleeves designated as types FT-96, FT-97, FT-98, and FT-472.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

VV-W-95	-	Wax, Paraffin, Technical.
PPP-B-566	-	Boxes, Folding, Paperboard.
PPP-B-636	-	Box, Fiberboard.
PPP-B-676	-	Boxes, Setup.
PPP-F-320	-	Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.
PPP-T-60	-	Tape: Pressure-Sensitive Adhesive, Waterproof, for Packaging.
PPP-T-76	-	Tape, Pressure-Sensitive Adhesive Paper, (for Carton Sealing).

MILITARY

MIL-P-116	-	Preservation, Methods of.
MIL-C-45662	-	Calibration System Requirements.

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-147	-	Palletized Unit Loads(40" X 48", 4-Way Partial and 4-Way Pallets).

FSC 5970

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM)

D149-61 - Dielectric Breakdown Voltage and Dielectric Strength of Electrical Insulating Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

3. REQUIREMENTS

3.1 Preproduction samples. This specification provides for preproduction inspection (see 4.3).

3.2 Material. The material for insulation sleeving shall be as specified herein. However, when a definite material is not specified, a material shall be used which will enable the insulation sleeving to meet the performance requirements of this specification. Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product.

3.2.1 Sleeving. Insulation sleeving types FT-96, FT-97, FT-98, and FT-472 shall be made of good quality, unbleached cotton yarn woven into uniform, circular braid.

3.2.2 Paraffin. The sleeves shall be thoroughly impregnated with paraffin (type I, grade AA of VV-W-95), the surplus paraffin removed, and the sleeves punched out to the internal diameter specified in table II. The amount of paraffin in treated sleeves shall not be less than 100 percent nor more than 150 percent of the weight of the unimpregnated sleeves.

3.3 Design and construction.

3.3.1 Length of insulation sleeving. The insulation sleeving shall be 3-1/4 ± 1/4-inches long.

3.3.2 Carriers, ends, size and picks per inch. The number of carriers, number of ends per carrier, size of cotton, and number of pick per inch for each type of insulation sleeving shall be in accordance with table I.

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TABLE I. Carriers, ends, size, and picks per inch.

Type	Nominal inside diameter (inch)	Minimum number of carriers	Ends per carrier	Size of yarn	Number of picks per inch
FT-472	3/32	28	1	26/2	33 ± 3
FT-96	1/8	32	1	26/2	30 ± 1
FT-97	5/32	44	1	24/2	32 ± 1
FT-98	1/4	44	2	26/2	23 ± 1

3.3.3 Internal diameter. The internal diameter of each type of insulation sleeving shall be such that it can be eased over a cylindrical rod of corresponding diameter as specified in the "Go" column of table II, but not large enough to be eased without forcing over a cylindrical rod of corresponding diameter as specified in the "No Go" column of table II. The cylindrical rods shall be smoothly tapered to a rounded point at the end which enters the sleeving.

TABLE II. Internal diameter of insulation sleeving.

Type	Nominal inside diameter (inch)	"Go" diameter (inch)	"No Go" diameter (inch)
FT-472	3/32	1/16	3/32
FT-96	1/8	3/32	1/8
FT-97	5/32	1/8	5/32
FT-98	1/4	7/32	1/4

3.3.4 Ends of insulation sleeving. The ends of each insulation sleeve shall be cut clean and approximately square and shall be free from fraying.

3.4 Dielectric withstanding voltage. When insulation sleeving is tested as specified in 4.4.3.3, they shall withstand 2,500 volts minimum average and 1,500 volts minimum individual without evidence of breakdown.

3.5 Workmanship. Insulation sleeving shall be processed in such a manner as to be uniform in quality and shall be free from defects that will affect life, serviceability, or appearance.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Test equipment and inspection facilities. Test equipment and inspection facilities shall be of sufficient accuracy, quality, and quantity to permit performance of the required inspection. The supplier shall establish calibration of inspection equipment to the satisfaction of the Government. Calibration of the standards which control the accuracy of inspection equipment shall comply with the requirements of MIL-C-45662.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
    (1) Inspection of product for delivery (see 4.4.1).
    (2) Inspection of preparation for delivery (see 4.4.4).

4.3 Preproduction inspection. Preproduction inspection shall be performed by the Government unless otherwise specified in the contract, after award of contract and prior to production, at a location acceptable to the Government.

4.3.1 Sample. 25 insulation sleeves of each type specified in the contract shall be submitted for preproduction inspection.

4.3.2 Test routine. Sample units shall be subjected to all the inspection procedures of this specification. Any rejects shall result in failure of preproduction sample.

4.4 Quality conformance inspection.

4.4.1 Inspection of product for delivery. Inspection of product for delivery shall consist of group A inspection (see table III and 4.4.2).

4.4.1.1 Inspection lot. An inspection lot, as far as practicable, shall consist of all insulation sleeves of the same type, produced under essentially the same conditions and offered for inspection at one time.

4.4.1.2 Rejected lots. If an inspection lot is rejected, the supplier may withdraw the lot, rework it to correct the defects, or screen out the defective units, as applicable, and reinspect. Such lots shall be separate from new lots, and shall be clearly identified as reinspected lots. Rejected lots shall be inspected using tightened inspection.

4.4.2 Group A inspection. Group A inspection shall consist of the examinations and tests specified in table III, and shall be made on the same set of sample units in any order which is satisfactory to the government inspector.

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4.4.2.1 Sampling plan. Statistical sampling and inspection shall be in accordance with MIL-STD-105 for general inspection level II. The acceptable quality levels (AQL) shall be as specified in table III. Major and minor defects shall be as defined in MIL-STD-105.

TABLE III. Group A inspection.

Examination or test	Requirement paragraph	Test paragraph	AQL	
			Major	Minor
Visual examination - - - - - - - -	3.2 thru 3.3.2, 3.3.4 and 3.5	4.4.3.1	1%	4%
Internal diameter - - - - - - - -	3.3.3	4.4.3.2	1%	N/A
Dielectric withstanding voltage-	3.4	4.4.3.3	1%	N/A

4.4.2.2 Disposition of sample units. Sample units which have passed all the group A inspection shall be delivered on the contract or order, if the lot is accepted.

4.4.3 Methods of examination and test.

4.4.3.1 Visual. Insulation sleeving shall be examined to verify that the design, construction, physical dimensions, and workmanship are in accordance with the applicable requirements of this specification.

4.4.3.2 Internal diameter. The internal diameter of insulation sleeves shall be tested for compliance with 3.3.3.

4.4.3.3 Dielectric withstanding voltage. Insulation sleeving shall be tested for dielectric withstanding voltage in accordance with ASTM D149-61, condition C-48/23/50 to comply with the requirements of 3.4.

4.4.4 Inspection of preparation for delivery. Sample items and packs shall be selected and inspected in accordance with MIL-P-116, table III, including rough handling tests, and the other specifications referenced herein, to verify conformance with the requirements in section 5 of this specification.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. Sleeves shall be cleaned in accordance with MIL-P-116, process C-1.

5.1.1.2 Drying. Sleeves shall be dried in accordance with one or more of the procedures listed in MIL-P-116. The procedure used shall not be injurious to the item.

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5.1.1.3 Preservative application. None required.

5.1.1.4 Unit packaging. One hundred sleeves shall be adequately packaged in accordance with MIL-P-116, method III, insuring compliance with the physical protection requirement (see 3.6 of MIL-P-116).

5.1.1.5 Intermediate packaging. Units of sleeves, 15 cubic inches and less, packaged as described in 5.1.1.4, shall be placed in intermediate containers conforming to PPP-B-566 or PPP-B-676. Intermediate containers shall be uniform in size and shape, shall be of minimum cube and tare, and shall contain multiples of five unit packages, not to exceed 100 packages or ten pounds. No intermediate packaging is required when the total quantity shipped to a single destination is less than 100 units.

5.1.2 Level C. Sleeves shall be preserved and packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity. This package may conform to the supplier's commercial practice for retail distribution when such meets the requirements of this level.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. The packaged item(s) shall be packed in fiberboard containers conforming to PPP-B-636, weather resistant, style optional, special requirement. In lieu of the closure and waterproof requirements in the appendix of PPP-B-636, closures and waterproofing shall be accomplished by sealing the center seams, ends, edges, and manufacturer's joints with waterproof tape, 2 inches wide, conforming to PPP-T-60, class 1 or PPP-T-76. Banding (reinforcement requirements) shall be applied in accordance with the appendix to PPP-B-636, using nonmetallic or tape banding only.

5.2.2 Level B. The packaged item(s) shall be packed in fiberboard containers conforming to PPP-B-636, class domestic, style and use requirements optional. Closure shall be in accordance with the appendix thereto.

5.2.3 Level C. The packaged item(s) shall be packed in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. This pack shall conform to the applicable carrier rules and regulations and may be the supplier's commercial practice when such conforms to the requirements of this level.

5.3 Unitized loads. Unitized loads may be utilized, at the option of the supplier, commensurate with the level of packing stated in the contract or order (see 6.1).

5.3.1 Level A. Insulating sleeving, shall be placed on a pallet, load type 1, conforming to MIL-STD-147. A fiberboard cap shall be employed over the load, having two sides extending down the stacked load at least 12 inches to accommodate marking requirements. The cap shall be PPP-F-320, class weather resistant, type SF, grade W5s fiberboard. The load shall be "bonded" to the pallet.

5.3.2 Level B. Insulating sleeving shall be packed as cited in level A, except fiberboard caps shall be class domestic.

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5.3.3 Level C. Insulation sleeving, shall be packed as cited in level A, except that pallets and caps shall be of the type, size and kind commonly used for the purpose and shall comply with the rules and regulations of the common carriers as applicable to the mode of transportation.

5.4 Marking (see 6.2). In addition to any special marking required by the contract or order, each unit package, intermediate and exterior containers shall be marked in accordance with MIL-STD-129.

5.5 General. Exterior containers shall be of a minimum tare and cube consistent with the protection required and shall contain equal quantities of identical items to the greatest extent possible.

6. NOTES

6.1 Intended use. Types FT-96, FT-97, FT-98 and FT-472 are intended for insulating individual wire splices when splicing communication cable.

6.2 Ordering data. Procurement documents shall specify the following:

(a) Title, number, and date of this specification.
(b) Levels of preservation and packaging and packing and applicable marking (see section 5).
(c) Samples required (see 3.1).

6.2.1 Indirect shipments. The preservation and packaging, packing, and marking specified herein apply only to direct purchases by or direct shipments to the Government and are not intended to apply to contracts or orders between the supplier and prime contractor.

6.3 Waiver of preproduction samples. When items for which preproduction samples are required, are currently in production on a government order, the contractor may apply to the contracting officer for waiver of preproduction samples on the new order. Any request for waiver should contain a statement which includes an indication that the specified samples have been submitted and approved on the current contract; that production of the item under the new order will be concurrent with, or follow without interruption, production on the current order; and that current production is in strict compliance with specified requirements for the new equipment; or describe any deviations authorized on the current order. Waiver of preproduction samples may be contingent on renegotiation of the contract to reflect the saving in cost to the contractor.

Custodians:
Army - EL
Navy -
Air Force - 11

Preparing activity:
Army - EL

(Project 5970-0143)

Review activities:
Army - EL
Navy -
Air Force - 11, 85

User activities:
Army -
Navy - CG, MC
Air Force - 17

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

a. GIVE PARAGRAPH NUMBER AND WORDING

b. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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